Holdouts, stability, and production decisions

> **Reading note.** This is an explanatory reconstruction of Jiaxuan Zou's literature synthesis. Published experiments belong to the cited teams. Numbered protocols and illustrative matrices are our planning synthesis unless explicitly attributed.

The blog's §10 treats validation as a set of different experiments. Predicting endpoint loss, remaining numerically stable, and achieving efficient production throughput each require evidence.

Validation sequence

1. **Extrapolation:** evaluate the frozen fit and recipe on larger sizes or longer horizons excluded from selection. Record error relative to the predeclared tolerance and seed variation.
2. **Combined recipe:** assemble winning data, architecture, and optimizer changes and compare against the baseline. Independent wins need not add; the combined recipe may require new tuning.
3. **Stability:** test learning rates above the intended operating point and long-horizon runs. Log gradient norms, update/weight ratios, nonfinite values, attention logits, and MoE load/drop statistics. The blog proposes LR multiples as stress tests; choose affordable magnitudes appropriate to the implementation.
4. **Implementation parity:** compare research and production updates on the same batch and state; then compare short trajectories and checkpoint resume. Define tolerances appropriate to precision and reduction order.
5. **Realized efficiency:** measure sustained tokens/second, peak memory, communication overhead, failures, and recovery. Match context length, batch, and parallelism assumptions.
6. **Dress rehearsal:** run the selected stack at an intermediate scale when the target is far beyond the evidence.

For a measured throughput r tokens/second on G accelerators, the idealized device-time estimate is GD/r. Add setup, evaluation, checkpoints, and recovery to obtain a usable budget. When reporting efficiency, specify whether r is aggregate throughput or per-device throughput.

Launch record

Gate	Evidence	Pass condition
Loss forecast	Untouched larger runs	Error within declared tolerance
Capabilities	Fixed downstream suite	Required domain floors
Stability	Stress and long-horizon logs	No unexplained numerical failures
Systems	Production-stack measurements	Fits memory and time budget

A holdout used to revise the fit or recipe becomes development data. Reserve a new holdout for the revised recipe. A failure should identify what was learned, which decision changed, and what fresh evidence is needed.

Sources: blog §10; Marin Delphi; Marin efficiency follow-up; OLMo 3, Appendix A.

Blog/notes crosswalk · Detailed companion